

Chain rule



- 1 Given that $y = u^3$ and $u = 2x + 3$, define y in terms of x .
- 2 Redefine the following functions as composite functions $f(u)$ and $u(x)$, where $u(x)$ is a polynomial:
 - a $f(x) = (5x^3 - 4x^2 + 3x - 5)^7$
 - b $f(x) = \sqrt[4]{2x^2 + 2x + 3}$
 - c $f(x) = \frac{1}{(4x^2 - 3x + 5)^3}$
- 3 To differentiate the function $y = (2x^4 + 6)^5$ using the substitution $u = 2x^4 + 6$:
 - a Find $\frac{du}{dx}$.
 - b Express y as a function of u .
 - c Find $\frac{dy}{du}$.
 - d Hence, find $\frac{dy}{dx}$.
- 4 For each of the following:
 - i Find $\frac{dy}{du}$.
 - ii Find $\frac{du}{dx}$.
 - iii Hence, find $\frac{dy}{dx}$.
 - a $y = (x + 5)^5$, where $y = u^5$ and $u = x + 5$.
 - b $y = (4x + 3)^{-1}$, where $y = u^{-1}$ and $u = 4x + 3$.
 - c $y = \sqrt{5 + x^2}$, where $y = \sqrt{u}$ and $u = 5 + x^2$.
 - d $y = \sqrt[3]{\frac{15}{x}}$, where $y = \sqrt[3]{u}$ and $u = \frac{15}{x}$.
- 5 Differentiate $y = (5x - 7)^2$:
 - a By expanding the brackets first.
 - b By using the chain rule.
- 6 Find the derivative of $y = \frac{1}{x^2 + 5}$. Give your answer in positive index form.
- 7 Find the derivative of $y = \sqrt{8x + 5}$ using the chain rule. Give your answer in surd form.

8 Differentiate the following using the chain rule



a $y = (4x + 3)^9$

c $y = (x^2 + x^{-3})^3$

e $y = (3x^2 - 4x + 2)^4$

g $y = \frac{(9x + 7)^{\frac{4}{3}}}{3}$

i $y = \frac{1}{x^4 - 4x^3 + 5x}$

k $y = \sqrt[5]{(4x + 1)^6}$

m $y = 5\sqrt{4 - \frac{1}{3}x}$

o $y = \left(\sqrt{x} + \frac{1}{\sqrt{x}}\right)^8$

q $y = -3\left(x + \frac{1}{x}\right)^5$

b $y = (2t^7 + 8t^3 + 3t + 5)^{-4}$

d $y = \sqrt[3]{x^2 - 5x}$

f $y = -3(3x + 4)^{10}$

h $y = \frac{1}{(x + 6)^5}$

j $y = \frac{2}{\sqrt{1 + x}}$

l $y = -4\left(\frac{1}{3}x + 1\right)^{-6}$

n $y = \frac{2}{1 - x\sqrt{5}}$

p $y = \frac{1}{\sqrt[3]{(9 - x)^4}}$

Gradients, tangents and normals

9 Consider the function $g(x) = (3 - x^5)^4$. Use the chain rule to evaluate $g'(-1)$.

10 Consider the function $g(x) = \sqrt{7 - 3x^2}$.

a Find $g'(1)$.

b Is the function increasing or decreasing at $x = 1$?

c Find the range of values of x for which the function is increasing or decreasing.

11 Consider the function $y = \sqrt{3 - 2x}$.

a Find the derivative of the function.

b To find where the tangent would be horizontal, what equation would we need to solve?

c Which of the following x -values has a tangent that is closest to horizontal?

A $-\frac{3}{2}$ B 10 C -10 D $\frac{3}{2}$

d Describe the behaviour of the gradient of the tangent as x decreases.

- 12 Find the gradient of $f(x) = (x - 5)^3$ at the point $(8, 27)$.
- 13 Find the gradient of the tangent to the curve $y = (3x - 1)^3$ at the point $(1, 8)$.
- 14 Find $f'(2)$ for the function $f(x) = 2(x^2 - 7)^5$.
- 15 Find the x -coordinate of the point at which $f(x) = (x - 2)^2$ has a gradient of 6.
- 16 Find the x -coordinate(s) of the point(s) at which $f(x) = (x + 2)^3$ has a gradient of 48.
- 17 Find the values of x where the tangent of $y = (x^2 - 1)^3$ is horizontal.
- 18 Find the values of x where the derivative of $y = (2x + x^2)^5$ is equal to zero.
- 19 The curve $y = \sqrt{x - 3}$ has a tangent with a gradient of $\frac{1}{2}$ at the point P . Find the coordinates of P .
- 20 For the function $f(x) = \frac{1}{2x - 7}$, find the values of x where $f'(x) = -\frac{2}{25}$.
- 21 Consider the semicircle defined as $y = \sqrt{169 - x^2}$.
- Find the derivative of the function.
 - Sketch the graph of the semicircle.
 - Find the equation of the tangent at the point $(5, 12)$.
- 22 Find the equation of the tangent to $y = (2x + 1)^4$ at the point where $x = -1$.
- 23 Find the equation of the tangent to $y = (2x - 1)^8$ at the point where $x = 1$.
- 24 Find the equation of the normal to $y = (3x - 4)^3$ at the point $(1, -1)$.
- 25 Find the equation of the normal to $y = (x^2 + 1)^4$ at the point where $x = 1$.
- 26 Consider the curve $f(x) = \frac{1}{2x + 3}$ at the point $(-1, 1)$.
- Find the equation of the tangent at the point.
 - Find the equation of the normal at the point.